

Class Meeting Handout #7

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Will review ever end in this course? The short answer is no. Today, we will review how structure affects acidity and introduce the central idea of structure-activity relationships that underpins this entire course.

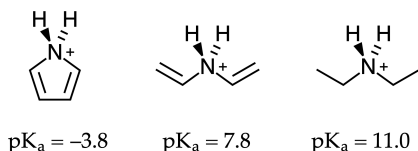
Structure Effects in Acidity

Closely related molecules can vary in acidity due to steric and electronic effects. Let us explore why acid strength (and base strength) vary with structure.

Review: Changing pK_a Values – 20 minutes

Structure affects reactivity. This is one of the core ideas of physical organic chemistry. Here we see clear examples of this as we explore how structure and environment affect acidity.

1. *para*-Nitrophenol is a useful leaving group for measuring rates of ester hydrolysis. It is a yellow colour when in its deprotonated form. I am performing an enzymatic hydrolysis of *p*-nitrophenylacetate as a probe for ester hydrolysis by a suite of hydrolases that are used in industry. I am doing the reaction at a pH value of 7.5. What percent of the nitrophenol is in the yellow coloured form? Explain why the pK_a value of *p*-nitrophenol is lower than the pK_a value for unsubstituted phenol.
2. Saccharine is (or used to be) present in soft drinks as its sodium salt. Can you show me the acidic proton of saccharine and propose a reason why it is likely never to be found in your coffee sweetener as anything other than a salt? Back up your answer with facts.
3. The pK_a value for several series of hydrides are presented in tables 2, 3, 4 and 5. Explain the trend in each series.
4. The pK_a values for protonated pyrrole, divinylamine and diethylamine are shown below. Explain the significant lowering of the pK_a value for divinylamine compared to diethylamine. Then explain the extreme pK_a value observed for pyrrole. How can closing the ring cause such a difference?



5. Explain the differences in pK_a values in water and DMSO as shown in table 6 on the other side of this sheet.

This document was produced using the $\LaTeX$ typesetting language with the Tufte-handout document class. Chemical diagrams were prepared in ChemDoodle,

Table 1: Selected pK_a values.↓

Acid	pK_a
Acetic acid	4.75
Phenol	9.95
<i>p</i> -Nitrophenol	7.10
Saccharine	1.60
Aniline	4.62
4-nitroaniline	1.11

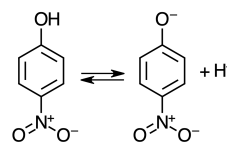


Figure 1: Acid equilibrium for nitrophenol.↑

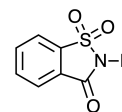


Figure 2: The structure of saccharine.↑

Table 2: pK_a values of 2nd row hydrides.↓

CH ₄	NH ₃	H ₂ O	HF
48	33	16	3

Table 3: pK_a values of halogen hydrides.↓

HF	HCl	HBr	HI
3	-6	-9	-10

Table 4: pK_a values of oxygen hydrides.↓

HClO	HClO ₂	HClO ₃	HClO ₄
7.5	-6	-9	-10

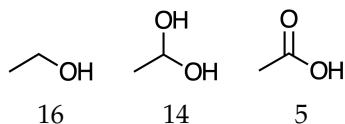
Table 5: pK_a values of carbon hydrides.↓

HC≡CH	H ₂ C=CH ₂	H ₃ C-CH ₃
25	44	50

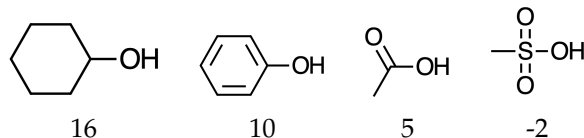
Exercise: More Changing pK_a Values – 20 minutes

Explain the relative pK_a values in the following series.

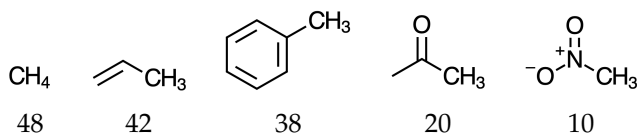
6. Explain the trend in the series below.



7. Explain the trend in the series below.



8. Explain the trend in the series below.



9. Explain the trend in the series below.

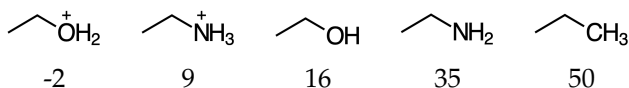


Table 6: Selected pK_a values.¹

Acid	pK_a in H_2O	pK_a in DMSO
HBr	-9	0.9
HCl	-8	1.8
HF	3.2	15
Picric acid	0.4	0
Acetic acid	4.75	12.3
Phenol	10.0	18.0
Methanol	15.5	29.0
Water	15.6	32
HCN	9.1	12.9
$\text{CH}_2(\text{CN})_2$	11.0	11.0

Learning Goals

There is more to a class meeting than what we do in 50 minutes of class time. After participating in the above exercises and completing the requirements described on the moodle site you will have achieved the following...

1. Understand the effect of resonance, induction, steric strain and electrostatic effects on the K_a value of an acidic functional group.
2. Be able to predict which of two related acids is stronger and explain why.
3. Understand the role of solvent in determining the K_a value of an acidic functional group.

Looking Ahead

Structure affects properties, acidity, reactivity, etc. How acidity is affected by the solvent or by changing substituents is pretty much the whole idea of this course. There will be much more of this. How can we measure and quantify these effects? Can we use them to investigate mechanism and transition state structure? Yes we can – in almost every class meeting from now on.